

## Examiner reports

### Q1.

This question discriminated well between lower and higher attaining students. The start of the question was accessible to most students. The later parts were more demanding and only the higher attaining students were successful.

Part (a) was completed successfully by a high proportion of students. Many selected a correct approach and manipulated the expression so that the binomial expansion could be

applied. The most common error was to extract a factor of 2 instead of  $\frac{1}{2}$  from the square root.

Those students who spotted the connection to the first part often did well on part (b). Full marks were available even if the first part of the question had gone wrong, provided  $-x^3$  was correctly substituted into their three-term expansion. Sign errors were the most frequently seen mistakes.

In part (c) students could still achieve credit even if their previous answers had gone wrong. Simply substituting their three-term answer to part (b) to form the integral would achieve the first mark.

Very few students demonstrated the insight required for (d)(i) and this question was intended to distinguish the most able students. Those who attempted an answer often talked about the shape of the curve or the use of trapezia. Those who realised that the key was that all the terms were positive often score full marks.

The idea being tested in (d)(ii) was standard, but it was made more difficult by the context. Students were not directed to find the range of validity for their expansion, but had to make the connection for themselves. The first mark could be achieved provided they used their expansion from part (b).

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Both parts of question (d) required explanation. Generally, this was not done well. Students needed to state a key fact which must be true/false and then explain the implications of this.

### Q2.

This question was answered correctly by a high proportion of students. Scripts showed evidence of algebraic integration from many students, whereas it had been expected that they would simply evaluate using allowed calculator functions. By far the most frequently chosen wrong answer was 60 and it is likely that these students evaluated one integral rather than splitting the area into two sections.

### Q8.

The use of the trapezium rule was well understood with a majority of students presenting

a fully correct solution to part (a) with the final answer given to the degree of accuracy requested. Students need to take extra care when transferring correct decimal values from a table into the formula; this was particularly relevant in this series to those values which had leading zeros, namely for  $f(4)$  and  $f(5)$ . The most common method error was to use end values 0 and 4 for  $x$  instead of the correct end values 1 and 5. A small minority of

students left their final answer in the form  $\frac{694}{1105}$  which resulted in the loss of the final accuracy mark. In part (b)(i), most students were able to integrate one of the terms

correctly but it was not uncommon to see the integral of  $x^{-\frac{3}{2}}$  resulting in a positive coefficient or an incorrect numerical simplification of  $1/-0.5$ . Very few students failed to score the method mark in part (b)(ii) for being able to deal with limits in a correct manner.

### Q9.

In part (a), apart from small number of students who did not differentiate the expression as

a product of two functions and obtained gradient functions such as  $\frac{dy}{dx} = -8x \sin 2x$ , hence gaining no marks, this question was usually answered very well. Some students

failed to simplify their expression for the gradient after substitution of  $\frac{\pi}{4}$ , thus giving rise to some very complicated forms for the equation of the line.

In part (b), almost all the students demonstrated they could integrate by parts, with many gaining full marks. Where marks were lost it was usually due to an incorrect coefficient

written in the integration of  $\frac{dv}{dx} = \cos 2x$  with  $v = 2\sin 2x$  being common. Some students failed to simplify their final expression and there were some cases where a non-exact decimal value was written as an answer. A small number of attempts at the volume instead of the area were seen.

### Q10.

In part (a), the integration proved difficult for many students who gave wrong coefficients

such as  $\frac{1}{2}$ , 2 or  $\frac{1}{1-2}$  and incorrect expressions such as  $-\frac{1}{1-2x} e^{1-2x}$ . The use of

substitution often did not lead to a correct result, with  $\frac{e^u}{u}$  being common. Many students found the elimination of  $\ln$  in  $e^{-2\ln 2}$  a challenge. Some students mistakenly thought that

changing their result into a decimal number and showing that it approximated  $\frac{3}{8}e$  was a sufficient proof.

In part (b), most students gained the first method mark for giving  $\frac{du}{dx} = \sec^2 x$  and the B mark for either changing the limits or using the original limits after their integration. The

replacement of  $dx$  with  $\frac{1}{\sec^2 x} du$  was sometimes seen for the first accuracy mark. The identity  $\sec^2 x = 1 + \tan^2 x$  was often seen but replacing  $\sec^2 x$  in terms of  $u$  often produced  $(1 + u)$  rather than  $(1 + u^2)$ .

Many students thought that they could integrate an expression that was a mixture of  $u$  and

$\tan x$  terms. A surprising number of able students had difficulty simplifying  $(1 + u^2)$

$$\sqrt{u} \text{ to } \left(u^{\frac{1}{2}} + u^{\frac{5}{2}}\right)$$

### Q11.

- (a) Those candidates who began by finding  $\frac{dy}{dx}$  usually found the correct gradient and most gave the equation of the tangent in the required form. A few made arithmetic errors such as  $y - 6 = 9x + 9$  leading to  $y = 9x + 3$ . A few, having found the gradient to be 9, then incorrectly used the negative reciprocal, i.e.  $-\frac{1}{9}$ . Some candidates did not realise that differentiation was required and attempted to find the gradient using the coordinates of the two points  $(-1, 6)$  and  $(2, 33)$ .
- (b) (i) Provided no arithmetic errors were made, particularly evaluating  $2^5 = 64$ , then the correct value of  $k$  was found from the equation of the curve. A few candidates used the tangent equation and although they found that  $k = 33$ , no credit was given.
- (ii) Those who used the equation of the line did not always earn the mark as a concluding statement such as “therefore lies on the tangent” was not made. Some worked backwards and found the gradient using  $(2, 33)$ . A few candidates confused the requests in parts (i) and (ii) and consequently earned no marks.
- (c) (i) The majority of candidates integrated correctly. Most substituted the limits in the correct order but many sign and arithmetic errors were made. A common error was evaluating  $\frac{5}{6} - \frac{54}{6}$  as  $\frac{49}{6}$  instead of  $-\frac{49}{6}$  and the correct value of the integral was rarely seen.
- (ii) Very few candidates found the area of a trapezium. Most found the area of a triangle instead. Those who used integration in order to find the area under the straight line were far more successful than those who tried to remember the basic formula from GCSE.

### Q12.

This question tested expansion of brackets and integration of negative exponents.

The majority of students used the binomial expansion or expanding brackets successfully in part (a). Typically in a(ii) they repeated their working from part (a)(i), replacing  $y$  with  $x^{-2}$ , rather than replacing it in their simplified expansion. Most students scored at least M1 for this part, however a significant number of students were unsuccessful in finding a correct expression for  $(2 - x^{-2})^3$  and hence did not find the correct values of  $p$  and  $q$ .

Those students who did not use “hence” but used a correct different method to the one they used in a(i), could gain a maximum of 2 out of the 3 marks. Part (b) made use of the answer to a(ii) and those that had found values for  $p$  and  $q$  were generally able to correctly integrate their expression,  $p + qx^{-4}$ , and evaluate it using the given limits in a

correct manner.

### Q13.

- (a) (i) The great majority of the students made progress here, most integrating correctly but some losing the final accuracy mark because of the omission of the constant of integration which was needed for the indefinite integral.
- (ii) There were three main approaches here and the most popular was to use  $u = \ln x$ ,  $\frac{du}{dx} = \frac{1}{x}$ . If the answer to part (i) was correct often all the marks were scored but sometimes there was a sign error in the final line. Using  $u = (\ln x)^2$ ,  $\frac{du}{dx} = 2 \ln x \cdot \frac{1}{x}$  was also successful for many students, although the omission of brackets on the first term lost the accuracy mark. Some students were not able to differentiate  $(\ln x)^2$  correctly. The third approach was using substitution for  $\ln x$ , but not many were able to follow that through and revert to a function of  $x$ . There were some poor attempts such as  $u$

$$= (\ln x)^2, \quad \frac{du}{dx} = \frac{1}{x^2} \quad \text{and} \quad \int (\ln x)^2 dx = \frac{(\ln x)^3}{3}.$$

- (b) Most students correctly made the initial step of finding  $\frac{du}{dx}$  correctly. A small number of students wisely set  $x = u^2$  and found  $\frac{du}{dx}$ . Writing the integrand in terms of  $u$  was found to be very challenging by many students and numerous attempts at falsely cancelling  $x^{\frac{1}{2}}$  prior to substitution often resulted in expressions that gained no further credit. Many who did get to the required expression failed to move on as they did not cancel by  $u$  and thought they could integrate to  $\ln(u^2 + u)$ . The more fundamental errors were  $\frac{u}{u^2 + u}$  becoming  $\frac{1}{u^2 + 1}$  or  $\frac{1}{u^2} + 1$ . Students should be made aware that the whole expression needs to stay within the integral; some had  $2u$  or the equivalent outside the integral. Some students who got to the last line used the  $x$  limits in the  $u$  expression.

### Q14.

After a successful start with the partial fractions many students went awry in later parts of the question, although fully correct solutions were seen from about 16% of the entry.

- (a) (i) The partial fractions were found correctly by the vast majority of students. Virtually all students knew what to do and had a correct opening line and only the occasional arithmetic error followed in the finding of  $A$  and  $B$ . The most popular method was to substitute  $x = -2$  and  $x = \frac{1}{3}$ , although some used simultaneous equations and others a mix of the two methods, often choosing to substitute  $x = 0$ .
- (ii) Again the vast majority of students had the correct form of the integrals but

many errors were made in the coefficients, particularly with the  $(1 - 3x)$  term,

often given simply as  $\ln(1 - 3x)$  or  $+\frac{1}{3} \ln(1 - 3x)$  and the  $(2 + x)$  term was

sometimes integrated  $\frac{3}{2} \ln(2 + x)$ . Most students went on to evaluate their integrals in the correct way, although it was not always clear how they used

the given limits and many made a sign error to obtain  $3 \ln 2 - \frac{1}{3} \ln 4$ , although  $\ln 1 = 0$  was well known. Many students who had the definite integral correct could not convert to the form of the requested answer. Several of those who

got as far as  $3 \ln 2 + \frac{2}{3} \ln 2$  did not see that all they needed to do was add 3 to  $\frac{2}{3}$ ; the answer  $\ln 8$  was common. The term rational may not have been sufficiently well known.

- (b) (i) It was expected that students would require little work to find the value of  $C$ ; it is immediate from the algebraic fraction by inspection from  $-6x^2 / -3x^2$ . However, relatively few noticed this. Many chose long division and found the correct value, but some used inefficient and time consuming methods based on undetermined coefficients. Some found  $C$  to a function of  $x$  although most gave  $C$  as a constant. Some students did not attempt this part of the question.
- (ii) All those who attempted this part knew that the requested area was given by integrating the function from part (b)(i), and most students recognised that this linked the parts of the question together. They gained credit if they had an exact answer to part (a)(ii) and added 2 to it, although many had a sign error when applying the limits and subtracted 2. Some students did not see the connection and tried to do the integral again, this time often going awry with answers such as  $\ln(2 - 5x - 3x^2)$ .

## Q15.

In part (a), differentiation had been well drilled and most candidates earned all 4 marks for the two derivatives. Occasionally, the  $+1$  was omitted or an extra 5 or  $+x$  was included, and a few felt the need to add  $+c$  to both their answers.

In part (b), those candidates who substituted  $x = -1$  into their expression for  $\frac{dy}{dx}$  often obtained a gradient of 12. However, not all used this value: some took the negative reciprocal and then formed the equation of the normal. Some candidates did not use their

expression for  $\frac{dy}{dx}$  in order to find the gradient but erroneously used the gradient of  $AB$ , with value 2, instead.

Surprisingly, a very small minority earned full marks in part (c). Most candidates verified the nature of the stationary point by considering the sign of the second derivative. However only a handful recognised the need to verify that the point was stationary in the first place, by showing that the first derivative was equal to zero.

In part (d)(i), most candidates were able to integrate the expression, with only the weakest candidates not making any attempt. Poor notation was used, with many including the integral sign after integrating. However dealing with the substitution of  $-1$  caused many errors. Manipulation of fractions caused problems too; it was disappointing to see many candidates with correct fractions being unable to combine these to give a value of 8 or equivalent. Many candidates did not find the actual value of the definite integral until part (d)(ii), and on this occasion full credit was given, but this may not be the case in future.

In part (d)(ii), most candidates calculated the area of the relevant triangle to be 2 and subtracted this from their answer to part (d)(i). Errors included writing the area of the

triangle  $\frac{4 \times 2}{2}$  as and careless arithmetic such as  $8 - 2 = 4$ .

### Q16.

This final question proved to be less demanding than has been the case in many of the past papers for this unit. Even the unstructured final part of the question produced a significant number of fully correct solutions from the candidates. Parts (a) and (b)(i) were found to be very straightforward with most candidates scoring heavily. It is worth recording that a number of candidates in part (b)(ii) failed to give sufficient justification for taking the gradient  $m$  to be  $-8$ . With the answer given, it was not acceptable to just state that the

gradient is  $-8$  without some clear indication that this value came from the value of  $\frac{dy}{dx}$  when  $x = 8$ . In part (c), the most common loss of a mark was through failing to simplify  $\frac{3}{\left(\frac{8}{3}\right)}$  correctly. The error of integrating  $x^{\frac{5}{3}}$  to get a term in the form  $kx^{\frac{7}{3}}$  was penalised more heavily.

In part (d), those who found the area of the triangle by solving the equations of  $OP$  and

$AP$  simultaneously to find the coordinates of  $P$  and then using  $A = \frac{1}{2}bh$  were generally much more successful than those who attempted to use integration, as their limits were frequently incorrect: 0 to 8 in both cases rather than 0 to 3.2 for  $OP$  and then 3.2 to 8 for  $AP$ . It was also surprising to see a significant minority of candidates finding the lengths of

$OP$  and  $AP$  and the angle  $OPA$  and then using  $\frac{1}{2}ab \sin C$  to find the area of the triangle; although a valid approach it took much more time and usually led to rounding errors. The method for finding the area under the curve was well understood but errors in part (c) frequently resulted in the area under the curve being greater than the area of the triangle: a clear error which should have been spotted by reference to the diagram.

### Q17.

In part (a), almost everyone earned the mark.

Part (b) was answered very well. Most candidates who realised the numerator was related to the derivative of the denominator and hence required a  $\ln$  function were successful in obtaining full marks. There were two very common errors:



- candidates losing the final accuracy by writing  $\frac{1}{6} \ln 91 - \frac{1}{6} \ln 21 = \frac{1}{6} \ln 70$
- candidates losing 2 marks by obtaining an incorrect value of  $k$ ; the most common value was 6 although  $\frac{1}{2}$  and  $\frac{1}{3}$  were also seen.

The major errors seen were expressions such as  $\frac{2x^2 - 1}{12x^2 - 6} \ln(4x^3 - 6x + 1)$ . Some candidates were able to recover by simplifying to the correct expression  $\frac{1}{6} \ln(4x^3 - 6x + 1)$  before substitution of the limits; however some candidates substituted for 3 and 2 through the whole unsimplified expression and therefore gained no credit.

### Q18.

In part (a), it was essential to use the quotient rule, as requested – it is specifically mentioned in the specification – and to show sufficient steps in the proof of the given answer. It was surprising that quite a number of candidates were unable to go from  $\frac{-\cos \theta}{\sin^2 \theta}$  to the final expression.

In part (b), it was good to see some completely correct solutions though some used  $30^\circ$  and  $45^\circ$ , which was incorrect. A substantial number of candidates who correctly started the question lost a minus sign in their work.

Many candidates fell at the first hurdle by not replacing  $dx$  by an expression in  $\theta$  and  $d\theta$ .

There were B marks available for a correct  $\frac{dx}{d\theta}$ , for  $\sqrt{\operatorname{cosec}^2 \theta - 1}$  converted to  $\sqrt{\cot^2 \theta}$  and for the limits in radians, but even these marks were often not earned. Unfortunately, many confused  $\frac{dx}{d\theta}$  with  $\frac{d\theta}{dx}$  and of course made no headway after that. Candidates who produced the correct numerical answer from their calculators without correct working did not gain marks.

### Q19.

In part (a), virtually all candidates showed confidence with finding the partial fractions and, bar a few slips, had this correct. Using two values of  $x$  to find  $A$  and  $B$  was by far the most popular method, although some candidates set up and solved simultaneous equations.

For part (b), there are several ways of finding the coefficients  $C$  and  $D$ , and these were all seen in candidates' work. The anticipated method of long division was seen less often than multiplying through by the denominator and then using two values of  $x$  or equating coefficients of powers of  $x$  or a mix of the two. Those who used long division sometimes failed to interpret their result. Using  $x = 0$  readily gives the value of  $D$ , but those who used

$x = \frac{3}{2}$  in finding  $C$  often made a mistake; the value of  $C$  is most readily found by

equating the coefficients of  $x^3$ . Those candidates who did not use an efficient method here would have wasted time that may have been more gainfully used in other questions.

A significant minority failed to make any progress in part (c) through not seeing the connection to parts (a) and (b). Most immediately went for an incorrect logarithmic integral and scored no further marks. Those who did attempt to use the results from parts (a) and (b) often made sign and / or coefficient errors during the integration. Some candidates who otherwise had this part all correct, failed to score the last mark because they made an error in attempting to manipulate their logarithms to the required form.

## Q20.

In part (a)(i), the coefficient of  $x$ , being an odd number, caused problems for some when completing the square; most students could not square 1.5 without a calculator; those who used fractions could not always simplify  $5 - \frac{9}{4}$ , and for those using decimals  $-1.25 + 5 = 3.75$  was commonly seen.

In part (a)(ii), the equation of the line of symmetry was not usually stated correctly, with several writing down the equation of a curve rather than a straight line; the incorrect equation  $y = \frac{3}{2}$  was seen far too often, whereas others simply wrote down numbers such as  $\frac{3}{2}$  or  $\frac{11}{4}$ ; this topic was not well understood by many students.

In part (b)(i), most attempts earned the method mark for eliminating  $y$  and collecting like terms, but, surprisingly, not all were able to solve the resulting quadratic equation. A common incorrect solution to  $x^2 - 4x = 0$  was  $x = \pm 2$ , whereas others obtained  $y = x^2 - 4x$  from the two equations and made no progress.

Integration of polynomials is well drilled and part (b)(ii) earned full marks for weak and strong students alike. There were few errors seen but the most common involved the constant term; some kept the term as 5 and others omitted the final term possibly confusing differentiation with integration.

In part (b)(iii), most students realised the need to substitute appropriate limits but, without a calculator, arithmetic errors were abundant;  $4^3$  was often evaluated as 48 and the

fractions and minus sign caused problems for many students who could not simplify  $\frac{64}{3} - 24 + 20$ .

Those who were unable to complete part (b)(i) often chose the value 5 as their upper limit, possibly since the coordinates of  $A$  were  $(0, 5)$ , but no credit was given for this. However, those who had the solution  $x = 2$  from their earlier work were given a method mark for using this value as their upper limit. A large number of students presented the value of the definite integral as their final answer to the area of the shaded region instead of considering the difference of the area of the appropriate trapezium and the value of the integral.

## Q21.

This question, which tested positive rational exponents and integration, was another good source of marks for many students, although for a large minority finding the square of  $x^{\frac{3}{2}}$



proved to be more of a challenge than was hoped, with  $x^{\frac{9}{4}}$  appearing far too often from otherwise promising students. The indefinite integration and evaluation of a corresponding definite integral were tackled with much more success, with many correct solutions seen.

## Q22.

In this question, students tended to do very well or very badly.

It was good to see that many students were able to complete this integration correctly. However there were many blank scripts, and the majority of students obtained only the B

mark for a correct answer for  $\frac{du}{dx}$ . Some students were able to make slight progress, and where M1 was obtained, many did manage to produce some form of  $\ln$  function, although many made a sign error and obtained  $\ln u - \frac{2}{u}$ . There were also many fully correct solutions seen.

## Q23.

In part (a)(i) candidates who recognised that the first step required was differentiation generally produced correct answers, but a few had trouble with the “ $-x$ ” term and others were unable to combine  $-1 + (-4)$  correctly. Some weaker candidates assumed that they could find the gradient using the coordinates of two points.

In part (a)(ii) once again there was much confusion between parallel and perpendicular lines, with many candidates giving the wrong gradient of the tangent. To earn the final mark it was essential to express the equation in the specified form and some failed to do so; for instance, an answer such as  $y + 5x = 17$  is not in the correct form.

In part (b)(i) the integration was generally correct and most candidates made an attempt to substitute the correct limits. A few however could not integrate “14” and left it as a constant. Very few were able to combine their fractions correctly and consequently only a small proportion of the candidates found the correct value of the integral.

In part (b)(ii) candidates who had only found the definite integral in part (b)(i) were generously rewarded with the remainder of the marks here. Not everyone recognised the need to find the area of a triangle in order to find the area of the shaded region. A few correctly found the area under the line  $y = 4x + 8$  by integration, but this is not the recommended approach.

## Q24.

Most candidates replaced  $\frac{8}{x^4}$  by  $8x^{-4}$  and differentiated correctly to score full marks for part (a).

Some other candidates failed to score the accuracy mark because they included a ‘+c’ in their

answer. A very large majority of candidates realised that the value of  $\frac{dy}{dx}$  at  $x = 1$  was required in

part (b), but a small minority did not appreciate that the value of  $y$  was also required. Although many candidates found the equation of the tangent correctly, it was not uncommon to see others finding the equation of the normal instead.

In part (c), a majority of candidates equated their expression for  $\frac{dy}{dx}$  to zero and attempted to solve the resulting equation, although only about half of the candidates obtained the correct coordinates of  $M$ . Those who had wrong powers of  $x$  in their answer to part (a) were restricted to just two marks in part (c). The most common incorrect methods seen either involved solving

$\frac{d^2y}{dx^2} > 0$  or looking for a link between the tangent in part (b) and the stationary point.

In part (d)(i), a high percentage of candidates were able to integrate  $\frac{8}{x^4}$  correctly, but errors in the integration of  $(x + 3)$  or the omission of the constant of integration resulted in a significant proportion of these candidates failing to score the final mark. Since answers to part (d)(ii) required explicit evidence that the answer to part (d)(i) had been used; the few candidates who just wrote down the answer without any other working gained no credit. However, most candidates applied the 'Hence', showing the values 2 and 1 substituted into their answer to part (d)(i) and carrying out the relevant subtraction. However, a significant number of candidates made arithmetical errors in carrying out the subtraction and so did not score the accuracy mark.

Only the more able candidates appreciated what was required in the final part of the question. The common wrong answers included 0 (presumably from the  $x$ -axis being a tangent), 12 (the  $y$  value from part (b)) and 5.5 (forgetting to consider the direction of the translation).

## Q25.

In part (a) most candidates were well drilled in integration and earned the first three marks; however, the fractions caused tremendous problems to most candidates who were simply

incapable of evaluating  $\left(\frac{1}{4} - \frac{2}{3} + 3\right) - \left(\frac{1}{4} + \frac{2}{3} - 3\right)$  without a calculator. Consequently, an incredibly small percentage obtained the correct value for the integral.

In (b) the area of the triangle was usually found to have value 2, although quite a few could not

simplify  $\frac{2 \times 2}{2}$ , and most candidates scored a mark for subtracting the area of the triangle from their answer to part (a). Only the best candidates found the correct value for the shaded region.

### Q26.

The majority of candidates answered part (a) successfully using the binomial expansion, Pascal's triangle or expanding brackets. Those candidates who wrote the expansion using the binomial coefficient notation from the formulae book did not always convince the examiner that they knew how to correctly evaluate the combinatorial coefficients.

In part (b)(i) very few candidates scored full marks with a large number of attempts seen where the numerator and denominator were integrated separately. Those candidates who realized the need for negative indices generally scored well. Most candidates picked up the method mark in the final part of the question for dealing with the limits in a correct manner. A few candidates just stated the correct answer for the definite integral without showing any working. Such an approach gained no credit as it did not explicitly indicate the use of 'Hence'.

### Q27.

Most candidates scored the marks for the two derivatives with relatively few omitting the minus signs. Some candidates managed to drop the 2 from  $\cos 2\theta$ . Many candidates brought the

commonly used parameter  $t$  into their expression using  $\frac{dx}{dt} = -\sin \theta$  or similar, but were not penalised. Virtually all knew how to use the chain rule, but many got mixed up with the coefficients in both expanding  $\sin 2\theta$  and simplifying the expression. The common error was to find  $k = 3$ . Some candidates got a result in terms of  $\sin \theta$  and some got no result at all, but any value of  $k$ , however found or invented, was allowed in part (a)(ii).

Part (a)(ii) was generally done well with most candidates demonstrating they both knew and could use the relation between the gradients of the tan gent and the normal, although a few

found an equation of the tan gent. The evaluation using  $\theta = \frac{\pi}{3}$  was generally accurately done; however candidates had to have derived  $k = 6$  correctly to gain the final mark for the equation of the normal.

In part (b) it was expected here that candidates would note the whole question was about the use of double angle formulae, and many did. Most got as far as at least trying to express  $\sin^2 x$  in term of  $\cos 2x$  and of those, many did it correctly, even if sometimes there was a mix of variables. The integration of  $\cos 2x$  was usually accurate, with the common error being to double  $\sin 2x$  rather than halve it. There were few sign errors. Virtually all candidates who achieved an

integral knew how to use the limits, but the real test here was in handling  $\sin 2\left(-\frac{\pi}{4}\right)$ ; many candidates lost credit through not showing their evaluation or getting the sign wrong.

However, many candidates, about 25% of the entry, went on to obtain the correct answer and give it in an exact form, although some did make a mistake with the signs, particularly the three minus signs involved with the lower limit. Of the few candidates who attempted the integral using parts most made little progress beyond the first line of working, not realising that in order to proceed with the integral they needed to change  $\cos^2 x$  back to  $\sin^2 x$ , and manipulate the resulting expression.

### Q28.

In part (a)(i), many candidates did not realise that differentiation was required in order to find the gradient of the curve, but instead erroneously used the coordinates of  $O$  and  $A$ . Some tried to rearrange the terms but usually made sign errors in doing so. In part (a)(ii), those who had the correct gradient in part (a)(i) were usually successful in finding the correct equation of the normal, though not everyone followed through to the required form, and sign errors were common. However, most obtained at least a method mark here, unless they found the equation of the tangent. The main casualties were once again those who always use the  $y = mx + c$  form for the equation of a straight line.

In part (b)(i), most candidates were well drilled in integration and scored full marks, although some wrote down terms with incorrect denominators. The limits 2 and 0 were usually substituted correctly, but it was incredible how many could not evaluate  $32 - 38 - 8$  without a calculator. The correct value of the integral was  $-14$ , but far too many thought that they had to change their answer to  $+14$  and so lost a mark. A small number of candidates differentiated or substituted into the expression for  $y$  rather than the integrated function.

In part (b)(ii), there was still some apparent confusion about area when a region lies below the  $x$ -axis. The area of the triangle was 6 units and hence the area of the shaded region was  $14 - 6 = 8$  but, not surprisingly, there were all kinds of combinations of positive and negative quantities seen here. It was worrying to see so many candidates failing to calculate the triangle area, with several finding the length of  $OA$  instead. Some able candidates found the equation of  $OA$  and the area under it by integration, but this was not the expected method.

### Q29.

Part (a)(i) Most candidates were able to integrate the expression with only the weakest candidates unable to do this basic integration. Poor notation was used with many including the integral sign after integrating. It would have been thought that this bad habit would have been corrected by the time of the examination. Many candidates did not find the actual value of the definite integral until part (a)(ii) and on this occasion full credit was given. It was alarming that many candidates who had correct fractions were unable to combine these to give a value of 8.4 or equivalent. Weaker candidates were seen

$$\frac{dy}{dx}$$
 substituting values into the expression for  $y$  or  $\frac{dy}{dx}$ , showing a complete lack of understanding.

Part (a)(ii) It was necessary to consider a rectangle of area 18 and then to subtract their answer from part (a)(i) in order to obtain the area of the shaded region. Many believed that the area of the rectangle was 9 and others failed to do this basic subtraction correctly, even when their answer to part (a)(i) was correct.

$$\frac{dy}{dx}$$
 Part (b)(i) Many candidates did not realise the need to find  $\frac{dy}{dx}$  before substituting the value  $x = 1$  and thus failed to score some easy marks for finding the gradient of the curve. A substantial number of candidates tried to calculate the gradient of the straight line between two points on the curve and scored no marks for this.

Part (b)(ii) Unfortunately many candidates tried to find the equation of the normal instead of the tangent to the curve. Otherwise, since there was a generous follow through in this

part of the question, most were able to score this final mark. The only exceptions were those who insisted on using  $y = mx + c$  where poor arithmetic often prevented them from finding a value for  $c$ .

### Q30.

The sign of  $p$  was often awry, although that of  $q$  was usually correct. This was not penalised again in part (b)(i) where a lack of simplifying the coefficients and signs in the expressions was a greater source of mark loss. It was also disappointing to see a

significant minority of candidates integrating the term  $+\frac{1}{x^6}$  instead of the printed term  $-\frac{1}{x^6}$ . However, in general, most candidates showed that they understood how to integrate  $x$  raised to a negative power.

Most knew how to evaluate a definite integral in part (b)(ii) but very few could maintain the numerical accuracy needed for the final mark. Those few candidates who just wrote down the value of the definite integral, presumably from a calculator, without showing how it had been obtained from the previous part (Hence) were awarded no credit. Some candidates, having correctly obtained  $-1.7$  for the value of the definite integral, went on to write 'Area = 1.7' as their final line of solution. This was not penalised and such candidates were awarded both marks.

### Q31.

- (a) This was well answered by the majority of candidates with many fully correct responses. The main error with the stretch was using the  $\wedge$ -direction with a scale factor of 2 and with the translation the vector often contained  $+1$ .
- (b) This part was well answered by many candidates.
- (c)
  - (i) This was not very well answered by weaker candidates with many just rearranging the expression and then writing down the answer given without any convincing step being shown.
  - (ii) This was answered better than part c(i) with many correct factorisations seen. The main error was with the rejection of the solution  $e^{2x} = -1$ .
- (d) Some candidates obtained the correct answer and earned full marks. Most candidates made a reasonable attempt at the integrals with reasonable success and most showed enough to earn the B mark for subtraction of their areas. Incorrect limits of 6 and 0 often spoiled what would have been very good attempts.

### Q32.

In part (a)(i) the majority of candidates realised the need to find the value of  $f(x)$  when  $x = 2$ . However, it was also necessary, after showing that  $f(2) = 0$ , to write a statement that the zero value implied that  $x - 2$  was a factor. It was good to see more candidates being aware of this.

In part (a)(ii), those who used inspection were the most successful here. Methods involving long division or equating coefficients usually contained algebraic errors.

In part (b)(i) a surprising number of candidates failed to see the need to differentiate in order to find the gradient at  $Q$ . Those who attempted to find  $\frac{dy}{dx}$  sometimes wrote it as  $3x^2 + x$ , but usually were aware of the need to substitute  $x = 2$ .

In part (b)(ii) those who had the correct gradient in part (b)(i) were usually successful in finding the correct equation of the tangent, and most obtained at least a method mark here.

In part (b)(iii) most were well drilled in integration and earned full marks, although some wrote  $\frac{x^4}{4} + \frac{x^2}{2} - 10$  and others gave  $\frac{x^4}{4} + \frac{x^2}{2} - \frac{10^2}{2}$  as their answer. For part (b)(iv) the correct limits were usually used, although many sign/arithmetic slips occurred after substitution of the numbers 0 and 2 and it was incredible how many could not evaluate  $6 - 20$  without a calculator. Very few candidates realised the need to show clearly that, although the integral from 0 to 2 gave a value of  $-14$ , the area of the shaded region was 14. A separate statement was needed and those who simply wrote  $4 + 2 - 20 = -14 = 14$  did not score full marks. Those more able candidates who made a statement about the region being entirely below the  $x$ -axis and who subsequently evaluated the integral from 2 to 0 correctly scored full marks.

### Q33.

This long calculus question with a final part covering transformations was a relatively high source of marks for many candidates. Most candidates were able to correctly differentiate

$2x^{\frac{3}{2}}$ , but not all evaluated it correctly at  $x = 4$ . The majority of candidates showed a good understanding of the method involved in finding the equation of the normal in part (a)(ii), although some others did not use the correct gradient. Although the vast majority of

candidates knew the rules for integrating  $8x^{\frac{1}{2}}$ , there was a significant number of such candidates who did not simplify the coefficient correctly.

It was not uncommon to see either ' $\frac{8}{3}x^{\frac{3}{2}}$ ' not simplified or simplified incorrectly to ' $12x^{\frac{3}{2}}$ '.

Part (b)(ii) was generally answered better than expected, with a majority of candidates applying a correct method for finding the area of the shaded region. Although many candidates realised that the transformation was a 'translation', there were frequently sign errors in the vector, or components were interchanged. It was also disappointing to see a significant increase in the number of descriptions that involved more than one transformation which, as usual, was given no marks; the two transformations quoted were translation and a stretch.

### Q34.

Those candidates who used long division in part (a)(i) scored no marks. Most candidates realised the need to show that  $p(-2) = 0$ , but quite a few omitted sufficient working such as  $p(-2) = -8 + 2 + 6 = 0$  together with a concluding statement about  $x + 2$  being a factor and therefore failed to score full marks.



Many candidates have become quite skilled at writing down the correct product of a linear and quadratic factor and these scored full marks in part (a)(ii). Others used long division effectively but lost a mark for failing to write  $p(x)$  in the required form. Others tried methods involving comparing coefficients, but often after several lines of working were unable to find the correct values of  $b$  and  $c$  because of poor algebraic manipulation.

In part (a)(iii), although many candidates tried to consider the value of the discriminant of their quadratic factor, quite a few used  $a = 1$ ,  $b = -1$  and  $c = 6$  (from the cubic equation) and scored no marks for this part of the question. Others drew a correct conclusion using the quadratic equation formula, indicating that it was not possible to find the square root of  $-8$  and others, after completing the square showed that the equation  $(x - 1)^2 = -2$  has no real solutions. Some wrongly concluded that because it was not possible to factorise their quadratic then the corresponding quadratic equation had no real roots.

Most obtained the correct  $y$ -coordinate of  $B$  in part (b)(i).

In part (b)(ii) it was pleasing to see most candidates being able to integrate correctly but a large number did not answer the question set and simply found the indefinite integral in this part. Many candidates use poor techniques when finding a definite integral and it was often difficult to see the evaluation of  $F(0) - F(-2)$  in their solution. Many obtained an answer of  $-10$  which was miraculously converted into  $+10$  with some comment about an area being positive. This and similar dubious working was penalized.

In part (b)(iii) some obtained an answer of  $-6$  for the area of the triangle by using  $-2$  as the base. Credit was given to candidates who later realised that the area of the triangle was actually 6. Unless candidates had scored full marks in part(ii) they were not able to score full marks in this part either, even if they obtained a correct value of 4 for the shaded area.

### Q35.

Most candidates gave the correct value for  $n$  in part (a). The most common wrong answer

was ' $n = -\frac{1}{4}$ '.

The expansion of  $\left(1 + \frac{3}{x^2}\right)^2$  in part (b) caused more difficulty than expected. A common wrong answer was ' $1 + 6x^{-2} + 9x^4$ ' although there were many other examples of incorrect answers offered.

In integrating  $\left(1 + \frac{3}{x^2}\right)^2$  in part (c) most candidates realised that their expansion in part (b) should be used but surprisingly the integration of 1 with respect to  $x$  caused just as many problems as the integration of the terms in  $x$  raised to negative powers. Although many candidates showed that they knew how to deal with the given limits, a lack of brackets in their numerical expressions frequently led to the wrong answer. The final mark was frequently lost because no **exact** answer (fraction or mixed number) was given.